

Web Application Project Proposal Revision 1

Title: Study Hub

Description

Study Hub is a full-stack web application designed for students to securely create, organize, and manage their personal study notes. Users can register for an account, log in, and create private notes containing a “Note Number, Title, Content, and Category”. Each note can be edited or deleted at any time, and only the logged-in user can access their own notes. The application also includes a movie search feature using a Movies API, allowing users to search for movies and view detailed information from their dashboard.

Problem or Need for application

Students need a secure and organized platform where they can store, manage, and access their study notes from anywhere. Study Hub provides students with a private note system while allowing them to organize notes by category and search for movie fans find information in one application. (Still to be added)

Core Features

- User registration, login, and logout.
- Create, view, edit, and delete personal study notes.
- Add a Note Number, Title, Content, and Category to each note.
- Create, edit, and delete note categories.
- Search movies using an external Movies API and view detailed movie information (Still to be added)

Current Application

Home Page

The home page provides users with access to the main features of Study Hub.

Features:

Application introduction.

- Register button for new users.
- Login button for existing users.
- Responsive layout for desktop, tablet, and mobile devices.

Registration

Users can create an account by entering:

- Name.
- Email address.
- Password.

Registration information is securely stored in MongoDB.

The registration page includes:

- Input validation.
- Error messages.
- Responsive form design.

Login

Users log in using:

- Email address.
- Password.

If incorrect login details are entered, the user receives an error popup:

- "Sign in failed. Check the details you provided are correct."

The login system includes:

- Secure authentication.
- User session management.
- Responsive layout.

Dashboard

After successful login, users access their personal dashboard.

The dashboard allows users to:

- Manage notes.
- Manage categories.
- Search movies.
- Logout securely.

Dashboard design:

- Black background.
- White menu text.
- Red Logout button with white text.

The dashboard supports:

- Desktop screens.
- Tablet screens.
- Mobile screens.

Notes System

Each note contains:

- Note Number.
- Title.
- Content.

Category.

Users can:

- Click the “Add Note” button to create a note.
- Select a category for the note.
- View saved notes.
- Edit notes.
- Delete notes.

Each note displays:

- Note Number.
- Title.
- Category.
- Content.
- Yellow Edit button.
- Red Delete button.

All notes are stored in MongoDB and linked to the logged-in user.

Users can only access their own notes

Category Management System (Still to be added.)

The Category page allows users to create and manage categories for organizing notes.

Each category contains:

- Category ID.
- Category Name.
- Category Description.

Example categories:

- Programming.
- Mathematics.
- Science.
- History.
- Personal Study.

Category Operations

Create

Users can:

- Open the Category page.
- Click "Add Category".
- Enter a category name and description.
- Save the new category.

Read

Users can:

- View all created categories.
- Select categories when creating or editing notes.
- Organize notes based on categories.

Update

Users can:

- Edit an existing category.
- Change the category name or description.
- Save updated category information.

Delete

Users can:

- Delete unused categories.
- Remove categories from their personal dashboard.

Movie Search Feature

The dashboard includes a movie search system connected to a Movies API still to be added.

Users can:

Search for movies.

View movie details including:

- Movie title.
- Description.
- Release date.
- Rating.
- Additional information provided by the API.

Technology Stack

- Framework: Next.js (App Router).
- Database: MongoDB.
- Authentication: Auth.js (NextAuth.js v5).
- Programming Language: TypeScript.
- Hosting Platform: Vercel.
- Version Control: GitHub.
- External API: Movies API.

CRUD Operations

Model 1: Notes

Create

Users can:

- Add a new note.
- Enter Note Number, Title, Content, and Category.
- Save the note.

Read

Users can:

- View all saved notes.
- View notes by category.
- Access only their own notes.

Update

Users can:

- Click the yellow “Edit” button.
- Update note information.
- Save changes.

Delete

Users can:

- Click the red “Delete” button.
- Remove notes from their account.

Model 2: Categories

Create

Users can:

- Add new categories.
- Give each category a name and description.

Read

Users can:

- View category lists.
- Select categories when creating notes.

Update

Users can:

- Edit category information.
- Update category names and descriptions.

Delete

Users can:

- Remove categories they no longer need.

Responsive Design

Study Hub is designed for multiple devices.

Desktop

Desktop users receive:

- Full dashboard navigation.
- Larger note and category cards.
- Side-by-side layouts.
- Full movie search results.

Tablet

Tablet users receive:

- Flexible layouts.
- Adjusted spacing.
- Touch-friendly buttons.
- Easy navigation.

Mobile

Mobile users receive:

- Column layouts.
- Mobile-friendly menus.
- Larger buttons.
- Vertical note and category displays.

Error Handling

Login Error

If your wrong password or login details are entered, the user receives:

“Sign in failed. Check the details you provided are correct.”

Additional error handling still to be added:

- Email already registered.
- Missing required fields.
- Invalid email format.
- Empty note information.
- Empty category information.
- Failed database requests.
- Failed movie API requests.

Loading States

Loading indicators during (To be added):

- Registration.
- Login.
- Creating notes.
- Editing notes.
- Deleting notes.
- Creating categories.
- Updating categories.
- Deleting categories.
- Searching movies.

Design System

Study Hub uses a consistent design style.

Main Theme

- Black background.
- White text.

Login Page

- Blue Login button.
- White button text.

Dashboard

- Black background.
- White menu text.
- Red Logout button.
- White logout button text.

Notes Section

- Black background.
- White text.
- Notes link with white text and white outline.
- Blue Add Note button with white text.
- Yellow Edit button with white text.
- Red Delete button with white text.

Category Section

- Black background.
- White text.
- Blue Add Category button with white text.
- Yellow Edit Category button with white text.
- Red Delete Category button with white text.

Reusable Components

- Navigation Bar.
- Authentication Forms.
- Buttons.
- Input Fields.
- Note Card.
- Category Card.
- Category Selector.
- Movie Card.
- Dashboard Layout.

Project Documentation

The README file (Information to be added):

- Project overview.
- Team members.
- Technology stack.
- Installation instructions.
- Environment setup.
- Database setup.
- API information to be added.
- Deployment instructions.
- Known issues, website still in revision 1 phase.
- Future improvements.

Final Summary

Study Hub is a full-stack student productivity application that provides:

- Secure user registration, login, and logout.
- Private user accounts.
- Notes with Note Number to be added, Title, Content, and Category.
- CRUD functionality.
- Dashboard management system.
- Movie search integration.
- MongoDB database storage.
- Next.js application architecture.
- Responsive design for desktop, tablet, and mobile.
- Consistent black and white design with blue, yellow, and red action buttons.
- Error handling with clear user feedback.